

DISCOVERY DAY 2025

REDEFINING POSSIBILITIES
WITH **SQL SERVER 2025,**
MICROSOFT FABRIC,
AND **AI**

12.06.2025 | SOFIA

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MICROSOFT FABRIC REAL-TIME INTELLIGENCE FOR DATA ENGINEERS

DISCOVERY DAY 2025



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azureplayer.net



[@KamilNowinski](https://www.youtube.com/@KamilNowinski)



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[linkedin.com/in/KamilNowinski](https://www.linkedin.com/in/KamilNowinski)

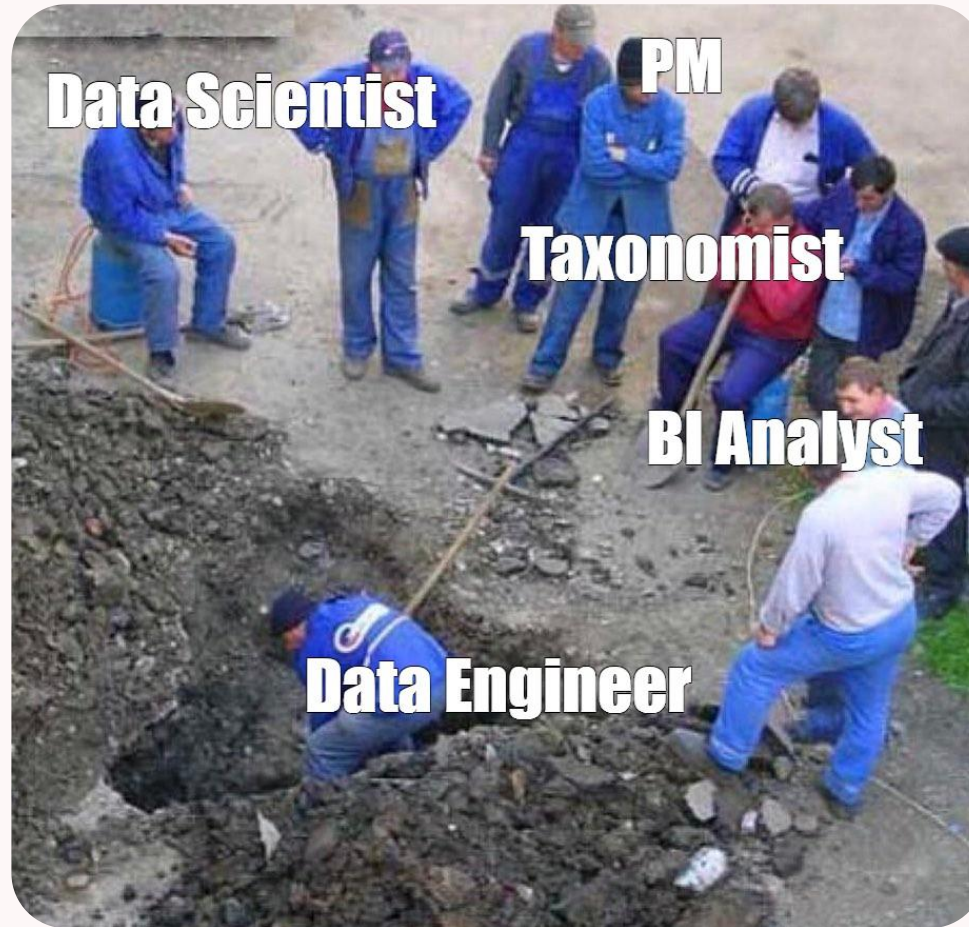
FAVORITE STUFF:



Agenda

- Introduction - data ingestion modes
- Overview of Fabric Real-Time Intelligence (RTI)
- Real-time data sources
- Fabric Eventstreams in use
- Selected unique capabilities of RTI
- Use cases for RTI
- Summary & resources

Data engineer's life



Success criteria



On-time data
delivery



Change
management

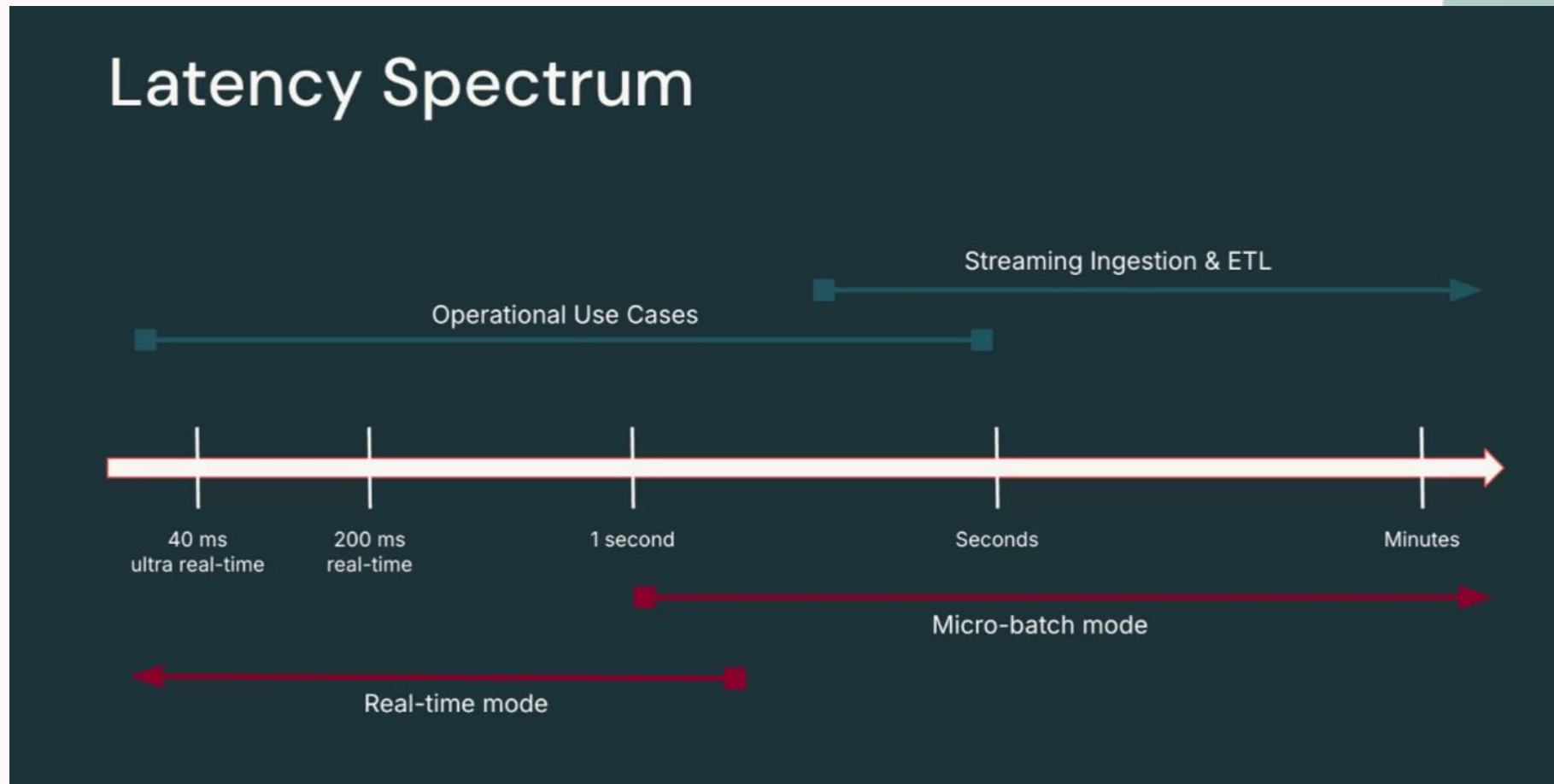


Good enough
data quality



Acceptable
cost

Data ingestion modes



Microsoft Fabric



Data
Factory



Data
Engineering



Data
Warehousing



Data
Science



Real-Time
Intelligence



Power BI



Databases



Industry
Solutions



AI



OneLake

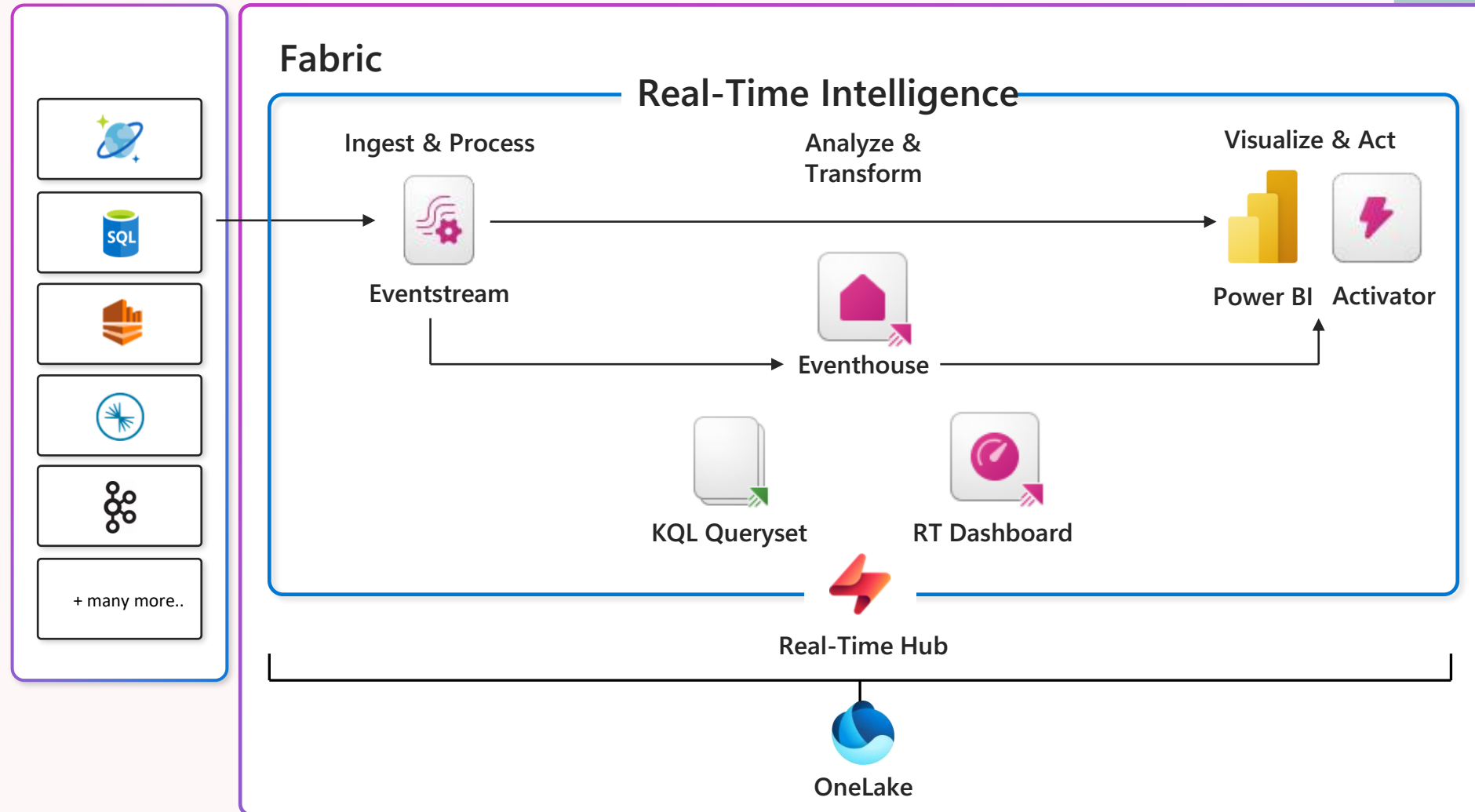


Purview



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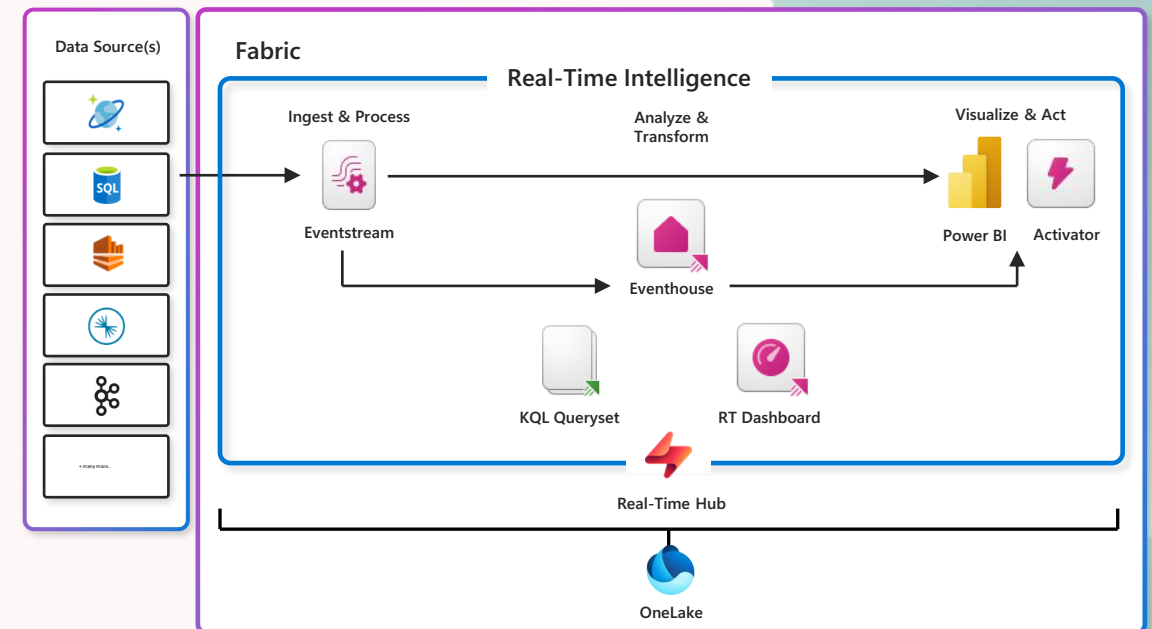
Overview of Fabric Real-Time Intelligence



Overview of Fabric Real-Time Intelligence

RTI item types

- **Eventstream:** Real-time data stream ingestion and transformation
- **Eventhouse:** Scalable storage and querying of real-time data
- **KQL Queryset:** A tool for creating and running queries in the KQL databases within an EventHouse
- **Real-Time Dashboard:** Interactive visualization and exploration of real-time data
- **Activator:** Automated actions triggered by real-time events



Overview of Fabric Real-Time Intelligence

Objects in KQL

- **Table:** Every data engineer knows what a table is, right? 😊
- **Function:** Built-in or user-defined function (stored or query-defined) for data transformation
- **View:** A virtual table (stored or query-defined) based on the KQL query result-set
- **Materialized view:** Aggregation query over a table or another materialized view, returning up-to-date result
- **Update policy:** An automation mechanism triggered when new data is written to a table

Data source



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DEMO

INTRODUCTION TO FABRIC REAL-TIME INTELLIGENCE

Real-time data sources

- Streaming cloud services
 - Azure: Event Hubs, IoT Hub
 - GCP: Pub/Sub
 - AWS: Kinesis
- Change Data Capture (CDC) streams from databases
- Apache Kafka (e.g. Confluent)

Eventstreams in use

1. Sources
2. Transformations
3. Destinations

The screenshot displays the Microsoft Fabric Eventstream interface. The top navigation bar includes the workspace name 'ES_DEMO', a security level indicator 'General (non-confidential)', a search bar, and a trial status 'Trial: 46 days left'. The left sidebar shows navigation options: Home, Create, Workspaces, Real-Time Intelligence, and a list of workspaces including 'EH_DEMO' and 'ES_DEMO'. The main canvas is in 'Edit mode' and shows a data pipeline. The pipeline starts with a source 'NYCTaxiStream' (labeled with a large '1'), which connects to an eventstream 'ES_DEMO-stream'. This stream then flows into a 'ManageFields' transformation (labeled with a large '2'), which maps 'VendorID' to 'VendorID' and 'passeng...' to 'passeng...'. The output of the transformation splits into two destinations: 'NYCTaxiEventhouse' and 'NYCTaxiLakehouse' (labeled with a large '3'). An 'Aggregate' transformation is also shown, calculating the 'Sum of passenger_count'. Below the canvas, the 'Test result' tab is active, showing a message about topology changes and a 'Refresh' button. The 'Data format' is set to 'Json' and the 'Time range' is '04/24/25 09:43:19 AM - 04/24/25 10:43:19 AM'. A table of test data is displayed below.

VendorID	tpep_pickup_datetime	tpep_dropoff_datetime	passenger_count	trip_distance
2	2022-06-01 00:33:18	2022-06-01 00:37:35	2.0	1.29
2	2022-06-01 00:22:49	2022-06-01 00:27:13	2.0	0.77
1	2022-06-01 00:55:17	2022-06-01 01:03:58	1.0	2.1
2	2022-06-01 00:52:55	2022-06-01 01:02:30	1.0	21.00

Eventstream sources and destinations

Sources	Destinations
• Azure Event Hub • Azure IoT Hub • Azure Event Grid Namespace • Azure Service Bus • Azure Data Explorer DB • Azure SQL DB Change Data Capture (CDC) • PostgreSQL DB CDC • MySQL DB CDC • Azure Cosmos DB CDC • Azure SQLMI DB CDC • SQL Server on VM DB CDC	• Google Cloud Pub/Sub • Amazon Kinesis Data Streams • Amazon MSK Kafka • Confluent Cloud Kafka • Apache Kafka • MQTT • Solace PubSub+ • Fabric Workspace Item events • Job events • OneLake events • Azure blob storage events • Custom endpoint • Sample data



Eventstream transformations

Transformations

- Filter
- Aggregate
- Union
- Expand
- Join
- Group by

Temporal Windows

- Tumbling
- Sliding
- Session
- Hopping
- Snapshot

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DEMO

LAMBDA ARCHITECTURE IN FABRIC
REAL-TIME ANALYTICS WITH REFERENCE DATA
FROM INSIGHTS TO ACTIONS WITH ACTIVATOR

Selected unique capabilities of RTI

Query acceleration over OneLake shortcuts

New shortcut

Review the folders and tables, and then Create. Each table and folder selected will appear as a unique shortcut in the location path.

Shortcut location

TPCHTest_Acceleration > Shortcut

Shortcut Name	Source	Accelerate (Preview) ⓘ	Status	Actions
 lineitem	TPCHLakehouse2 > Tables > lineitem	☒	Yet to create	 

lineitem data policies

Query acceleration ⓘ

☒ On

Caching period (days) ⓘ

365

Min 1, max 36500 days

- High-performance queries over shortcuts in Eventhouse
- Policy on top of external delta tables - number of days to cache data

```
.show external table lineitems operations query_acceleration statistics
```

- Multiple sources supported
 - OneLake shortcuts (+all destinations supported by OneLake shortcuts), Azure Data Lake Store Gen1, Amazon S3, Google Cloud Services, Azure Blob Storage
- See performance benchmark: <https://dcode.bi/blog/accelerate/>

Selected unique capabilities of RTI

Fabric Events

- Events based on Fabric **monitor activities**, **OneLake operations** or actions on Fabric items
- Can be configured in **Real-Time hub**
- Alerts (activators) can be set to **trigger actions** based on events of specific types (e.g. new file added in specific lakehouse)
- **An action can be running a Fabric item**

Set alert

Monitor

Source
OneLake events

Hide event types
Microsoft.Fabric.OneLake.FileCreated

Show applied filters

Condition

Check
On each event

Action

☐ Send me an email

☐ Message me in Teams

☒ Run a Fabric item

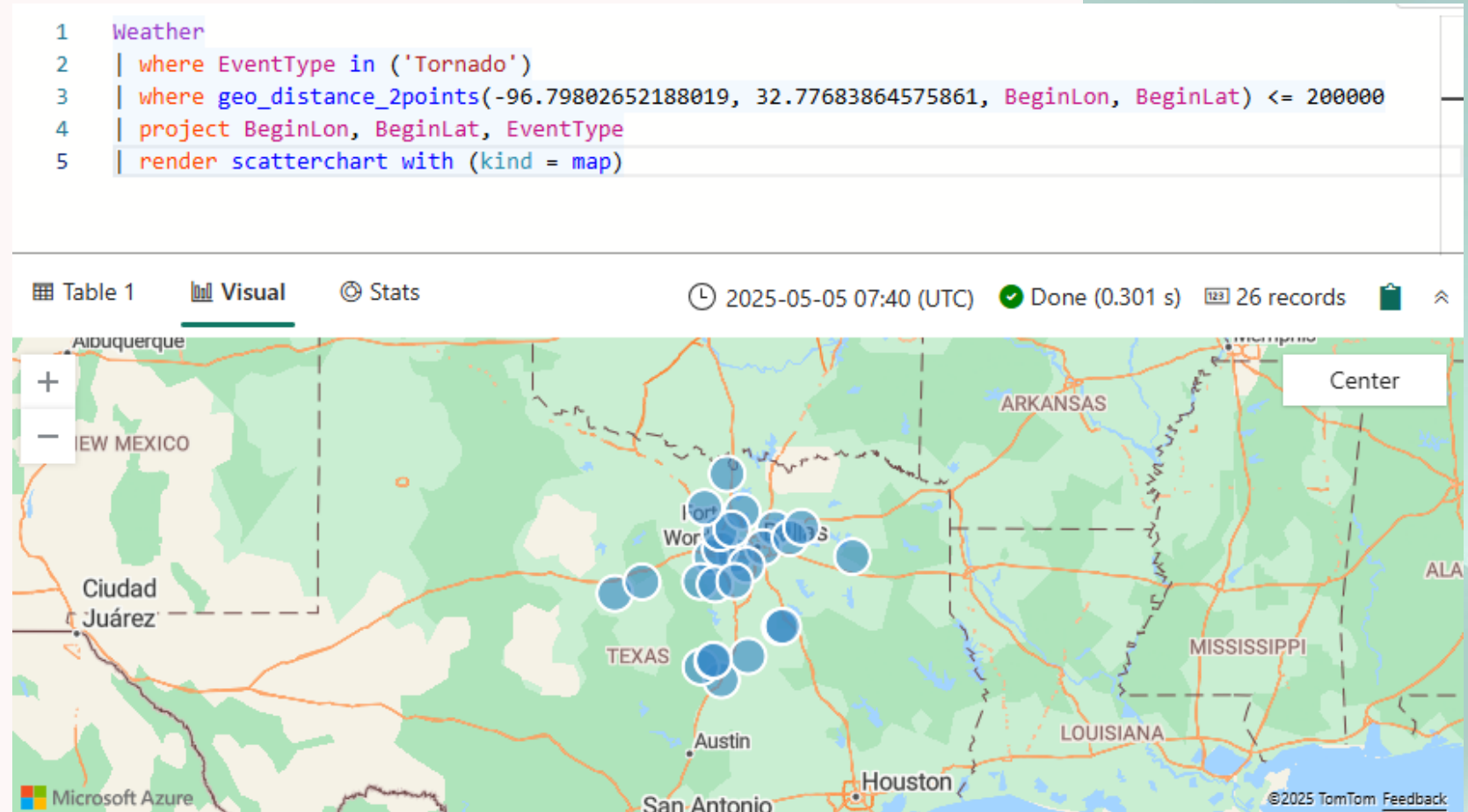
Workspace
(0) Load NB

Item
NB Load

Selected unique capabilities of RTI

Geospatial analysis

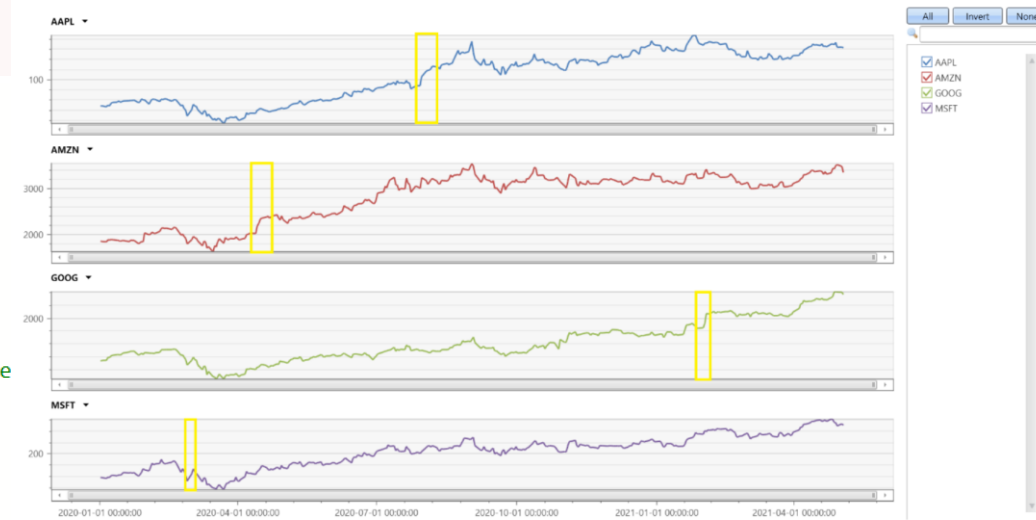
- Functions
- Visualizations
- Clustering
- Joins



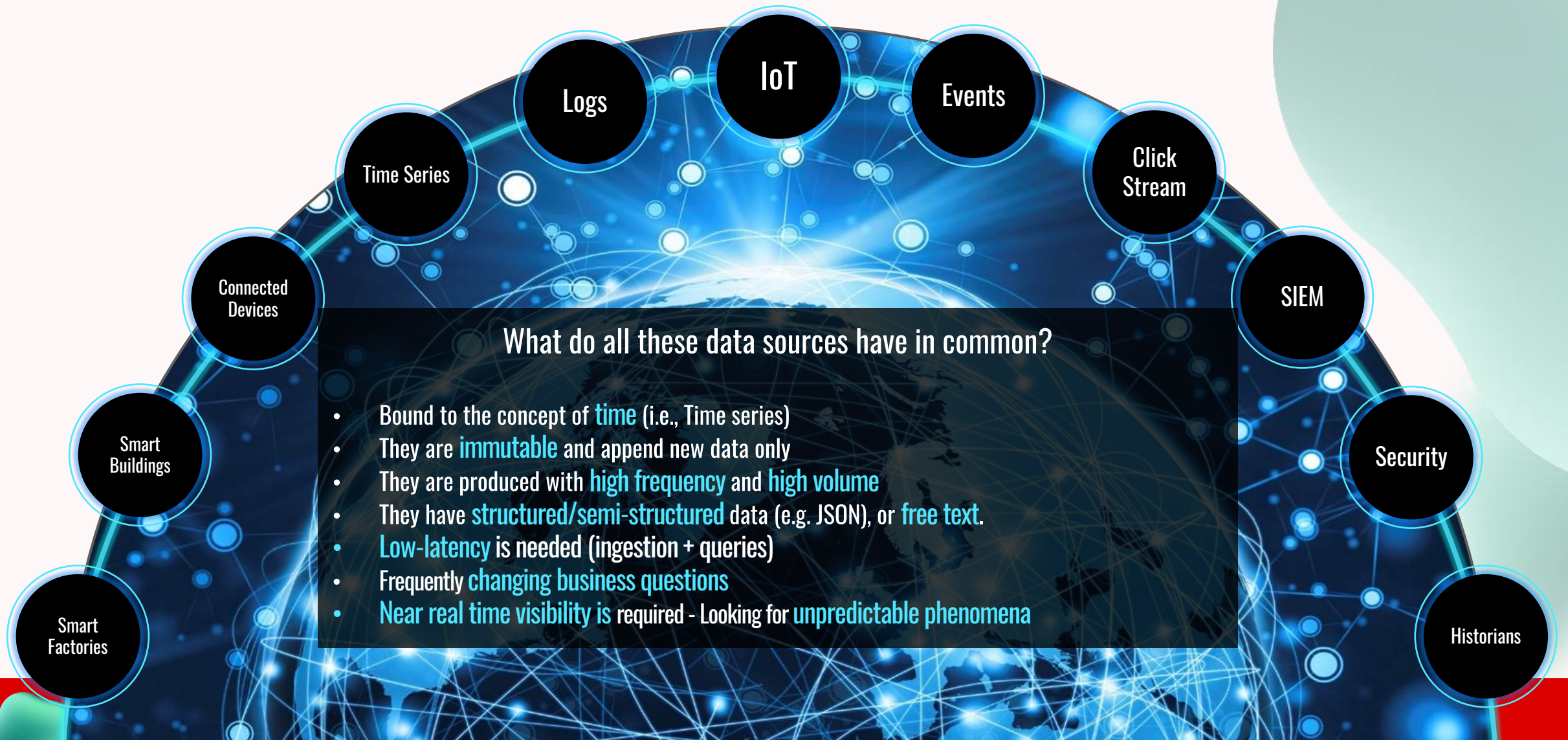
Selected unique capabilities of RTI

KQL built-in functions

```
1 TopStocksCleaned
2 | where DateTime >= datetime(2019-01-01) and Ticker in('AAPL', 'AMZN', 'GOOG', 'MSFT')
3 | partition by Ticker
4 (
5     order by DateTime asc
6     | extend pDate=prev(DateTime), pAdjClose=prev(AdjClose)
7     | extend delta = AdjClose - pAdjClose
8     | scan with_match_id=m_id declare(down:bool=false, step:string) with
9     (
10         // if state of s1 is empty we require price increase, else continue as long as price doesn't decrease
11         step s1: delta >= 0 and (delta > 0 or isnnull(s1.delta)) => step = 's1';
12         // exit the 'rally' when price decrease, also forcing a single match
13         step s2: delta < 0 and s2.down == false => down = true, step = 's2';
14     )
15 )
16 | where step == 's1' // select only records with price increase
17 | summarize (start_date, start_AdjClose, start_delta)=arg_min(pDate, pAdjClose, delta), end_date=max(DateTime), trading_days=count(), total_delta=sum(delta) by Ticker, m_id
18 | extend delta_pct = total_delta*100.0/start_AdjClose
19 | summarize arg_max(delta_pct, *) by Ticker
20 | project Ticker, start_date, end_date, trading_days, delta_pct, start_AdjClose, total_delta
21 | order by delta_pct
```



Use cases



Summary

- Always define what „real-time” means
- Fabric RTI provide capabilities for data engineers to handle real-time data
- Verify whether your data characteristics fit RTI
- Fabric RTI is NOT a replacement for Data Warehouse or Lakehouse

Resources

Resource	Hyperlink
Fabric RTI documentation	https://learn.microsoft.com/en-us/fabric/real-time-intelligence/
Fabric RTI end-to-end tutorial	https://learn.microsoft.com/en-us/fabric/real-time-intelligence/tutorial-introduction
Medallion architecture in Fabric RTI	https://blog.fabric.microsoft.com/en-us/blog/21597?ft=All
Kusto Detective Agency	https://detective.kusto.io/
Brian Bonk's blog:	https://dcode.bi/blog/
KQL.how:	https://kql.how/



THANK YOU!
We'd love to hear your thoughts!

Please take a minute to complete short feedback survey.



QR CODE

Time to indulge in something sweet
— enjoy the incredible gelato by:

